

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (ISE) Degree Examinations
 December 2024

IS2101-1 – COMPUTER ORGANIZATION & DESIGN
 (For IS)

Duration: 3 Hours

Max. Marks: 100h

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. _____, _____ are 2 parts of instruction.

A) Memory and data	B) opcode and operand
C) Cache memory and register	D) ALU and CU
2. **Fetch phase and _____ are 2 phases of instruction execution.**

A) Execution phase	B) Decode phase
C) Load phase	D) Transfer phase
3. **In single precision number _____ bits are used for representing mantissa**

A) 8 Bits	B) 16 Bits
C) 4 Bits	D) 23 Bits
4. **Three types of buses are data bus, _____, control bus**

A) Address bus	B) Bi-directional bus
C) Free bus	D) Uni-directional bus
5. **2's Complement of -6 is _____**

A) 1010	B) 1110
C) 0110	D) -0110
6. **In sequential circuit binary Multiplication _____ is scrolled right to achieve multiplication.**

A) Accumulator -> Carry -> Multiplier	B) Multiplier -> Carry -> Accumulator
C) Carry -> Accumulator -> Multiplier	D) Carry -> Accumulator -> Multiplicand
7. **_____ holds address of next instruction to be executed.**

A) MAR	B) PC
C) MDR	D) IR
8. **The usual BUS structure used to connect the I/O devices is _____**

A) Star BUS structure	B) Single Bus structure
C) Multiple bus structure	D) CACHE
9. **The method of synchronizing the processor with the I/O device in which the device sends a signal when it is ready is?**

A) Exceptions	B) Signal Handling
C) DMA	D) Interrupts
10. **Micro program is**

A) The name of a source program in micro computer	B) set of micro-instructions that defines the individual operations in response to a machine-language instruction
C) a primitive form of macros used in assembly language programming	D) a very small segment of machine code

11. To overcome the lag in the operating speeds of the I/O device and the processor we use
 A) Buffer Spaces B) Interrupt Signals
 C) Status Flags D) Exceptions
12. The method which offers higher speeds of I/O transfers is
 A) Interrupts B) Memory mapping
 C) Program-controlled I/O D) DMA
13. Which of the following is not true about DMA?
 A) DMA is an approach of performing data transfers in bulk between memory and the external device without the intervention of the processor. B) The DMA controller acts as a processor for DMA transfers and overlooks the entire process.
 C) The DMA controller has 3 registers. D) DMA is a Disk Memory Accumulator
14. _____ BUS arbitration approach uses the involvement of the processor.
 A) Centralised arbitration B) Distributed arbitration
 C) Random arbitration D) Fixed arbitration
15. The return address from the interrupt-service routine is stored on the
 A) System heap B) Processor register
 C) Processor stack D) Hard disk
16. The smallest entity of memory is called _____
 A) Cell B) Block
 C) File D) Instance
17. The instruction -> Add LOCA, R0 does _____
 A) Adds the value of LOCA to R0 and stores in the temp register B) Adds the value of R0 to the address of LOCA
 C) Adds the values of both LOCA and R0 and stores it in R0 D) Adds the value of LOCA with a value in accumulator and stores it in R0
18. The small extremely fast, RAM's are called as _____
 A) Cache B) Heaps
 C) Accumulator D) Stacks
19. The primary memory (also called main memory) of a personal computer consists of
 A) RAM only B) ROM only
 C) Both RAM and ROM D) Cache
20. Which of the following statements are not correct about the main memory of a computer?
 A) In main memory, data gets lost when power is switched off. B) Main memory is faster than secondary memory but slower than registers.
 C) They are made up of semiconductors. D) Main memory is slower than secondary memory.

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With a neat diagram explain the connection between processor and memory.	6	L*2	1	1
	b) Design the Sequential multiplier circuit. Perform the multiplication of 1110 (multiplicand) and 0001 (multiplier) using sequential multiplier.	10	L3	2	2
2.	a) Write the basic operational steps involved in execution of the following instructions. i. Move R1,R2 ii. SUB R0,LOCA	6	L1	1	1
	b) State and explain an algorithm of restoring division. Divide 11 by 3 using this method.	10	L3	2	2
3.	a) Explain the computational details of multiplying two 4-bit numbers (M) 5 and (Q) -13 using Booth's algorithm. Compare the result obtained with fast multiplication bit-pair recoding method.	8	L3	2	2

- b) Explain the algorithm of non-restoring division of integer with an example.

8 L2 2 1

Unit – II

4. a) With a neat diagram explain the three bus organization of data-path and write a control sequence for the instruction MUL R1, R2, R3.

8 L3 3 2

- b) Write a note on

i. Interrupt Hardware

ii. Daisy Chaining to handle the interrupt requests

8 L2 4 1

5. a) Design a single-bus organization of data-path inside a processor. Write the control sequence for execution of the instruction Add (R1), R2.

8 L3 3 2

- b) With a neat diagram explain the basic organization of micro-programmed control unit. Also write the micro-routine for the conditional branch instruction BR<0.

8 L2 3 1

6. a) What is micro-programmed control and micro-routine? For the below shown control sequence, write the corresponding control words (CWs).

i. PCout, MARin, Read, Select4, Add, Zin

ii. Zout, PCin, Yin, WMFC

iii. MDRout, IRin

iv. R3out, MARin, Read

v. R1out, Yin, WMFC

vi. MDRout, SelectY, Add, Zin

vii. Zout, R1in, End

8 L3 3 2

- b) Explain in detail about DMA transfer. Analyze the registers involved in DMA transfer.

8 L2 4 1

Unit – III

7. a) With block diagram explain associative and set associative cache mapping functions.

10 L2 5 1

- b) Write a note on ROM.

6 L1 5 1

8. a) Explain with a neat diagram how SRAM can be implanted. Analyze the read and write operations in SRAM.

6 L1 5 1

- b) Write a note on Virtual memory organization. Explain how virtual addresses are translated into physical addresses.

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
